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10. Oxygen forms several molecular compounds of formula X_2O. These include compounds with fluorine and chlorine.

(a) Some thermodynamic data for F_2O and Cl_2O are given below.

Substance	Standard enthalpy change of formation, $\Delta_f H^\theta$ / kJ mol^{-1}	Standard molar entropy, S^θ / J K^{-1} mol^{-1}
F_2O	24.5	247
Cl_2O	80.3	266
O_2		205
F_2		203
Cl_2	0	

A student examines the data and states that there is sufficient information to find the minimum temperature for decomposition of F_2O to its elements but not for Cl_2O.

Is the student correct? Justify your answer. [3]

(b) F_2O is a strong oxidising agent and can oxidise Xe to XeF_4 whilst releasing oxygen gas.

(i) Write an equation for this process. [1]

(ii) Show that the xenon is oxidised in the process. [1]

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(c) F_2O reacts slowly with water to produce HF, a weak acid.

$$F_2O + H_2O \longrightarrow 2HF + O_2$$

A small amount of F_2O is added to $400\,cm^3$ of water at 15 °C and it reacts completely. The reaction produces $82\,cm^3$ of oxygen gas at this temperature and 1 atm pressure.

(i) Calculate the concentration of the HF solution formed. [2]

concentration = ... $mol\,dm^{-3}$

(ii) Calculate the pH of the HF solution formed. [3]

$$K_a\,(HF) = 6.6 \times 10^{-4}\,mol\,dm^{-3}$$

pH = ...

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(d) Cl_2O can react with water in a reversible reaction.

$$Cl_2O + H_2O \rightleftharpoons 2\,HOCl$$

This reaction can occur in non-aqueous solvents and in a particular solvent the value of the equilibrium constant, K_c, at 273 K is 5.0 and it has no units.

(i) Write an expression for the equilibrium constant K_c. [1]

(ii) A student studies the equilibrium at a higher temperature. She adds samples of 0.40 mol of H_2O and 0.14 mol of Cl_2O to the non-aqueous solvent giving a total volume of 1000 cm^3.

At equilibrium the concentration of HOCl is 0.22 mol dm^{-3}.

I. Find the value of the equilibrium constant K_c at this temperature. [3]

K_c = ...

II. State, giving a reason, whether this reaction is endothermic or exothermic. [2]

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III. HOCl can act as an oxidising agent.

$$2HOCl\,(aq) \;+\; 2H^{+} \;+\; 2e^{-} \rightleftharpoons Cl_2\,(g) \;+\; 2H_2O\,(l) \qquad E^{\theta} = +1.63\ V$$

Give the formulae of all the metal ions in the table that can be oxidised by HOCl (aq). Explain why you have chosen these ions. [2]

	Standard electrode potential, E_{θ}/V
$Fe^{3+} + e^{-} \rightleftharpoons Fe^{2+}$	+0.77
$Tl^{3+} + 2e^{-} \rightleftharpoons Tl^{+}$	+1.25
$Pb^{4+} + 2e^{-} \rightleftharpoons Pb^{2+}$	+1.69
$Co^{3+} + e^{-} \rightleftharpoons Co^{2+}$	+1.82

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